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PIXE Simulations for Material Characterization: From Pure Elements to Complex Alloys

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Proton-Induced X-ray Emission (PIXE)

1. Theoretical Background

The physical principle underlying the simulations in this study is Proton-Induced X-ray Emission (PIXE). When a material is bombarded by MeV-range protons, these projectiles interact electromagnetically with target atoms. Through Coulomb scattering, an incident proton (p^+) can eject an inner-shell electron (e^-) from a target atom (A), creating a vacancy and leaving the atom in an excited ionized state (A^{+*}):



To restore the ground state, an outer-shell electron almost instantaneously transitions downwards to fill the hole. This relaxation releases excess energy as a radiative X-ray photon (γ_X), exactly equal to the energy difference between the initial and final electron states:



Since these quantized energy levels are uniquely defined for every chemical element, the emitted X-rays act as an elemental “fingerprint”, allowing for the unambiguous identification of the target’s composition. To theoretically validate these results, the simulated emission energies are compared against the Evaluated Atomic Data Library (EADL). Transitions are identified using the Siegbahn notation, which classifies X-rays by the shell where the initial vacancy occurs (K, L, M) followed by a Greek letter (α, β, γ) representing the relative energy and distance of the shell from which the filling electron originates.

2. Simulated Materials and Spectra Analysis

This section presents the various targets analyzed through PIXE simulations, ranging from pure elements to complex alloys. All simulations were performed using a 10 MeV proton beam, with the exception of the pure silver target, which was utilized to evaluate the emission yield dependence across an energy range of 5 MeV to 50 MeV.

2.1. Pure Silver (Ag)

The initial analysis was conducted on a pure silver target ($Z = 47$) with a mass density of 10.50 g/cm^3 . Using the standard 10 MeV proton beam, the simulated PIXE spectrum clearly exhibits the characteristic X-ray signatures of silver. To focus on the most significant data, the plotted energy range was restricted to the high-energy region (21–26 keV); transitions below this threshold were omitted since their emission probabilities were extremely low, resulting in no discernible peaks in the simulation output.

As expected from the EADL database, the simulated silver spectrum is dominated by the high-intensity K_α doublet and the secondary K_β multiplet, as we can see in Fig 1.

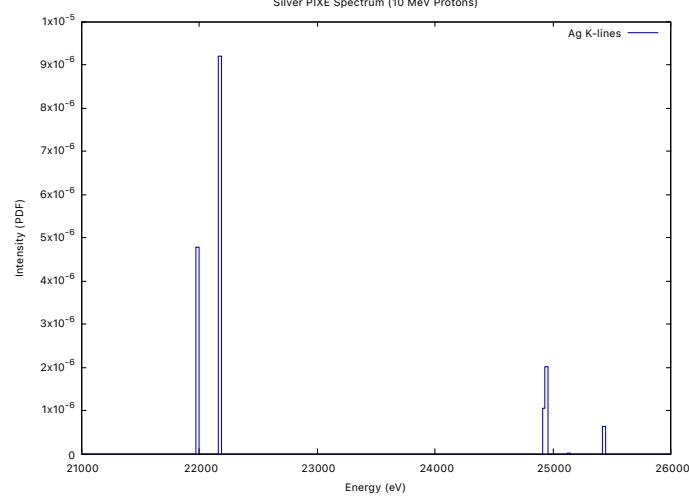


Figure 1: Simulated PIXE spectrum of pure silver (Ag) irradiated with a 10 MeV proton beam, highlighting the K_α and K_β groups.

A quantitative comparison between the theoretical transition energies and the simulated peaks is summarized in the table below.

Transition	Theory Energy (eV)	Sim. Energy (eV)
K L3 ($K_{\alpha 1}$)	22139.0	~ 22140
K L2 ($K_{\alpha 2}$)	21960.90	~ 21960
K M3 ($K_{\beta 1}$)	24920.50	~ 24920
K M2 ($K_{\beta 3}$)	24888.80	~ 24890
K N2,3 ($K_{\beta 2}$)	25420.50	~ 25450

Energy dependence. To investigate the dependence of X-ray emission on the projectile's kinetic energy, the same silver target was subsequently bombarded with proton beams ranging from 5 MeV to 50 MeV.

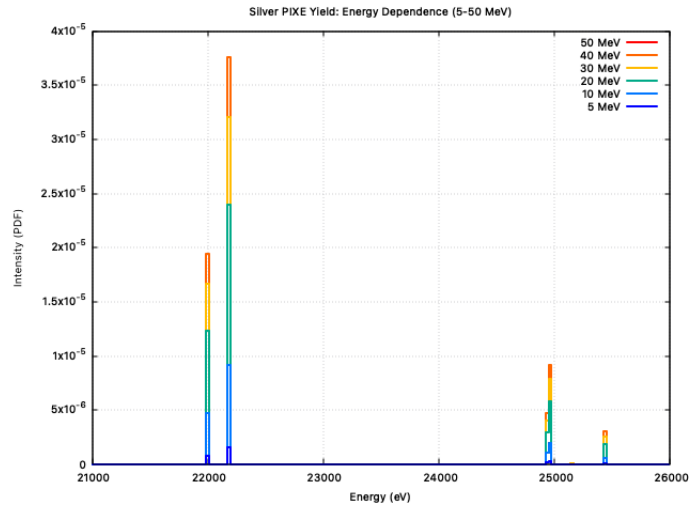


Figure 2: Simulated PIXE spectra of a pure Silver target at different incident proton energies.

The experimental results shown in Figure 2 clearly demonstrate that the intensity of the X-ray peaks is highly dependent on the energy of the incident proton beam. Physically, this is

due to the ionization probability: a faster, more energetic proton has a much higher chance of colliding with and ejecting an inner-shell electron compared to a slower one.

As the beam energy increases, the number of vacancies created in the K-shell grows significantly, leading to a much higher number of X-ray emissions. It is important to note that while the intensity changes, the position of the peaks on the energy axis remains perfectly constant, confirming that the emission energy is a "fingerprint" of the atom itself and does not depend on how the initial vacancy was created.

2.2. Bronze (Cu-Sn Alloy)

Following the analysis of pure silver, a bronze alloy (Cu-Sn) was modeled to characterize the X-ray emission spectrum in the case of a multi-elemental matrix. The material, defined with a mass density of 8.0 g/cm³, represents a typical binary alloy where Copper acts as the matrix and Tin as the alloying element, as presented in the table below.

Element	Atomic Number (Z)	Mass Percentage (%)
Copper (Cu)	29	86.9%
Tin (Sn)	50	13.1%

The resulting simulated spectrum for the bronze alloy is shown in Figures 3 and 4. As expected for a multi-element mixture, the spectrum displays the superposition of the characteristic X-ray emission lines of its main components. Due to its predominant concentration, the Copper K-series dominates the spectrum, while the Tin peaks are less intense, reflecting its lower atomic fraction in the alloy.

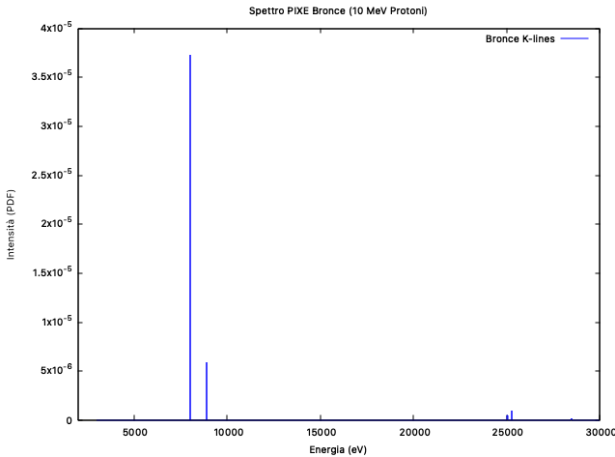


Figure 3: Full spectral overview of the Bronze.

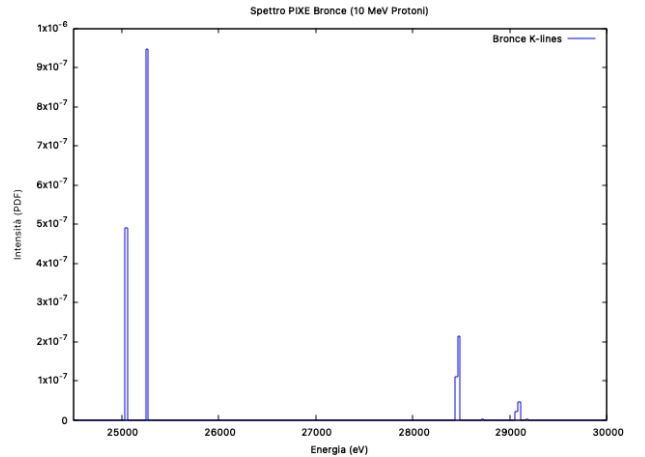


Figure 4: Detailed zoom of the Tin peaks.

For clarity, the main characteristic transitions contributing to the spectrum are summarized in the table below.

Element	Transition	Theory Energy (eV)	Sim. Energy (eV)
Cu	K L3 ($K_{\alpha 1}$)	8005.71	~ 8050
	K L2 ($K_{\alpha 2}$)	7984.67	~ 8030
	K M3 ($K_{\beta 1}$)	8865.34	~ 8910
Sn	K L3 ($K_{\alpha 1}$)	25257.20	~ 25270
	K L2 ($K_{\alpha 2}$)	25023.20	~ 25040
	K M3 ($K_{\beta 1}$)	28473.70	~ 28490
	K N2 ($K_{\beta 2}$)	29080.40	~ 29085

As expected for a multi-element mixture, the plot clearly displays the superposition of the characteristic lines of its components. Due to its predominant concentration, the spectrum's intensity is dominated by the Copper K-series (~ 8.0 keV), while the Tin peaks (~ 25.2 keV) are visibly less intense, reflecting its lower atomic percentage in the mixture. This separation of energies allows an accurate elemental analysis of the alloy's components.

2.3. Eurofer97 Steel

The study continues with the analysis of Eurofer97. This material was defined in the simulation with a mass density of 7.80 g/cm^3 , and its complex multi-elemental composition is summarized in the table below.

Element	Symbol (Z)	Mass Percentage (%)
Iron	Fe (26)	88.35%
Chromium	Cr (24)	8.58%
Others*	W, Mn, V, C, Ta, S	3.07%

The resulting PIXE spectrum for Eurofer97 is shown in Figure 5. To better resolve the multiple contributions of the alloying elements, the analysis focuses on the 5–8 keV energy range.

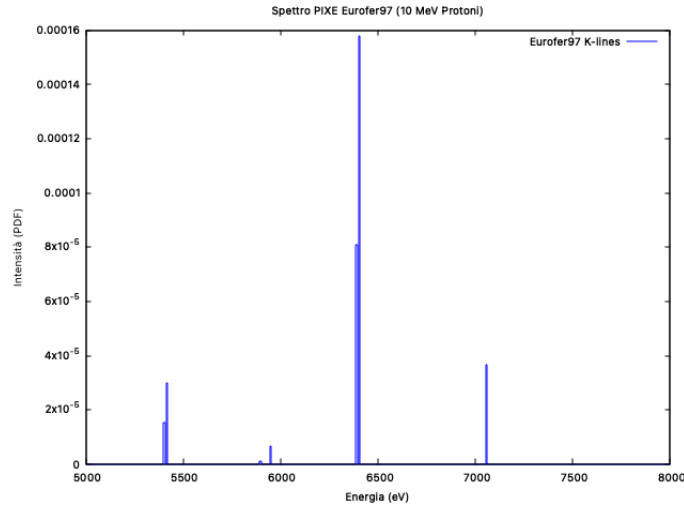


Figure 5: Simulated PIXE spectrum of Eurofer97 (10 MeV protons) focused on the 5–8 keV region.

The spectral analysis confirms a high degree of correlation with the theoretical emission energies. The main characteristic peaks identified in the Eurofer97 spectrum are summarized in the table below.

Element	Transition	Theory Energy (eV)	Sim. Energy (eV)
Cr	K L3 ($K_{\alpha 1}$)	5376.79	~ 5410
Mn	K L3 ($K_{\alpha 1}$)	5858.68	~ 5900
Fe	K L3 ($K_{\alpha 1}$)	6362.71	~ 6400
Fe	K M3 ($K_{\beta 1}$)	7016.95	~ 7060

As demonstrated by the data, the PIXE simulation successfully resolves the dominant contributions of the Iron-Chromium matrix, while also proving sensitive enough to detect trace elements like Manganese, despite its low mass concentration (0.43%).

2.4. White Gold Alloy (Au-Ag-Pd)

The final case study focuses on white gold, with a density of 13.0 g/cm^3 . Unlike the previous steel sample, this alloy is composed exclusively of heavy noble metals. The material was defined in the simulation with the elemental composition summarized in the table below.

Element	Symbol (Z)	Mass Percentage (%)
Gold	Au (79)	58%
Silver	Ag (47)	32.20%
Palladium	Pd (46)	9.50%

The simulated PIXE spectrum of the white gold target is presented in Figure 6. The spectral landscape is characterized by two distinct groups of peaks: the L-series transitions of Gold in the lower energy range (9–14 keV), and the K-series emissions of Silver and Palladium in the higher energy region (21–25 keV).

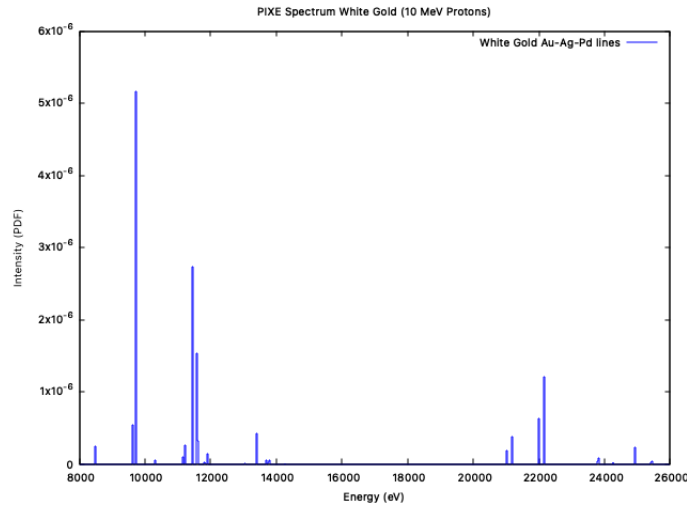


Figure 6: Simulated PIXE spectrum of White Gold (10 MeV protons).

The peaks observed in the spectrum are in excellent agreement with the expected characteristic X-ray energies for the alloy’s components, as summarized in the table below.

Element	Transition	Theory Energy (eV)	Sim. Energy (eV)
Au	L3 M5 ($L_{\alpha 1}$)	9709.80	~ 9710
Au	L2 M4 ($L_{\beta 1}$)	11475.30	~ 11440
Au	L2 N4 ($L_{\gamma 1}$)	13422.50	~ 13380
Pd	K L3 ($K_{\alpha 1}$)	21150.40	~ 21180
Ag	K L3 ($K_{\alpha 1}$)	22139.0	~ 22160
Ag	K M3 ($K_{\beta 1}$)	24920.50	~ 24940

As expected, the high atomic number of Gold ($Z = 79$) results in its K -series transitions being absent from the recorded energy range. Since the K -shell binding energy of Gold is approximately 80.7 keV, its ionization cross-section for a 10 MeV proton beam is negligible compared to that of its L -shell ($\sim 11.9 \text{ keV}$). Consequently, the L_{α} and L_{β} lines serve as the necessary and reliable fingerprints for Gold in this analytical context. Furthermore, the spectrum highlights the quantitative accuracy of the simulation: since Silver ($Z = 47$) and Palladium ($Z = 46$) possess almost identical ionization cross-sections, their observed K_{α} peak intensity ratio (~ 3.4) perfectly matches their nominal mass fraction ratio ($32.20\%/9.50\% \approx 3.4$).